



Measuring net energy use

As you know better than we do, some carbon removal approaches can require a good deal of energy. This raises questions about whether the approach removes carbon on net, or whether it takes away clean energy from decarbonization efforts more broadly. To address these concerns, we put together [a set of principles](#) after conversations with buyers, registries, and energy modeling experts to help CDR companies choose the best possible sources of energy for their operation. These principles include guidance for how to calculate the amount of clean energy to procure to counterbalance some or all of a company's grid electricity emissions.

Recently, we partnered with [WattTime](#), who analyzes electricity usage for real-time, marginal and average grid emissions data, to make it easier to access the data required to run the calculations on a facility's emissions impact and to support better energy and siting choices. Using this program is optional, but we hope you'll take advantage, since it's a great opportunity for free insights!

This document contains:

- How to measure net energy use and emissions for CDR facilities
- How to access average and marginal hourly emissions data via WattTime

How to calculate

Primer on calculating facility energy emissions

For CDR suppliers looking to counterbalance some or all of their grid electricity emissions by procuring clean energy, Frontier recommends that CDR suppliers near-term pursue one of two approaches:

Approach	Definition	What this looks like for CDR projects
Temporal matching	Match the use of a facility's grid electricity with the generation of procured clean energy on an hourly basis	• Projects procure hourly energy attribute certificates (EACs) from clean energy projects located within the same grid subregion or balancing authority to match some or all of their hourly grid electricity use. • Projects calculate the emissions from any unmatched or partially matched hours using hourly average emissions factors from the relevant grid boundary and subtract them from net project CDR.



Annual volumetric matching with emissions screen	Procure clean energy procured that matches the grid electricity used by the project with an emissions screen	- Projects procure a volume of clean energy that matches or exceeds their facility's annual grid energy use (volumetric matching). - Projects also calculate an <u>over-procurement factor</u> that ensures the volume of clean energy procured effectively counterbalances the emissions impact of the facility's grid electricity use today. Detailed guidance here
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Example calculations

Below is an overview of how to calculate facility emissions using each approach, with an example calculation provided in the workbook [here](#):

	Temporal matching	Volumetric matching with an emissions screen
Facility energy use and emissions	Estimate hourly facility electricity use over the course of the year and calculate emissions using average hourly emissions factors ¹ .	Calculate grid hourly emissions factors for past three years: - Pull marginal operating and build hourly emissions factors² - Pull average hourly emissions factors Calculate facility emissions using hourly emissions factors: $e = \sum(f_t \cdot ef_t)$ where f_t is the expected facility energy use for hour t and ef_t is the emissions factor for hour t. Determine the annual average expected facility emissions based on the past three years of emissions factors.
Clean energy procurement	Determine which hours it makes financial sense to purchase hourly EACs for new clean energy for. Ensure purchasing is additional and meets the criteria here . For any unmatched hours, use hourly average emissions factors to reduce facility net CDR. For example, for an unmatched hour where the emissions factor = 0.2 tCO₂e/MWh and facility energy use is 2 MWh/ton, net CDR for that hour would be reduced by 40% (0.2*2).	Calculate emissions impact using hourly emissions factors for clean energy procurement, assuming 100% annual volumetric matching: $a = \sum(g_t \cdot ef_t)$ where g_t is the expected clean energy generation for hour t and ef_t is the emissions factor for hour t. Determine the annual average expected avoided emissions based on the past three years of emissions factors. Calculate a procurement factor (p) based on the delta between facility and clean energy emissions: $p = \frac{e}{a}$. Note that $p = 1$ is the minimum allowable procurement factor. - Do this calculation for both average hourly and marginal hourly emissions factors, and choose the larger of the two procurement factors. Ensure purchasing is additional and meets the criteria here . ³
Notes	This is relatively straightforward to implement in cases where there is a liquid market for hourly-matched EACs, but could be more difficult for smaller loads today.	

¹ Details on how to access below.

² Use a 50/50 weighting between operating and build marginal emissions factors.

³ Any shortfall in annual PPA production should be subtracted from project net CDR. For example, if a project uses 100 GWh in a given year but the PPA only produces 95 GWh due to climate conditions, the emissions associated with the 5 GWh of unmatched grid electricity use should be counted against project CDR.



Using WattTime data

What data is available

Three years of hourly marginal and average emissions data

- Initially for 2022 - 2024, refreshed annually
- Available across the globe: eGRID subregion-level granularity for the US, balancing authority-level granularity for Canada and Australia, and country granularity for all other non-US countries.

How to access

To access Marginal Operating Emissions Rate (MOER) hourly emissions data:

- Go to <https://gridemissionsdata.io/>
- Register for data access
- Select the region(s) on the map the correspond with your project(s), and click 'download'

To access Marginal Build Emissions Rate (MBER) hourly emissions data:

- Go to <https://www.gem.wiki/MBERs>
- Click 'hourly download' and select the desired BA/ISO (US) or country (non-US)

To access Average Operational Emissions Rate (AOER) data:

- Email chris@watttime.org and include the location you are requesting data for. Locational granularity should be lat/lon or similar for US and Canada, and country level elsewhere.
- WattTime will send an NDA for signature
- Upon signing, WattTime will share the data for the specified location via CSV

Tips on using WattTime data

- MOER, MBER, and AOER units are lbs of CO₂/MWh
- Timestamps are UTC , period ending

Questions?

- On Frontier's energy expectations & energy calculations: Zeke (hausfath@stripe.com)
- On data access and interpretation: Chris Patino (chris@watttime.org) and Henry Richardson (henry@watttime.org)